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# Regulation of cancer cell signaling pathways by mushrooms and their bioactive molecules: Overview of the journey from benchtop to clinical trials

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ABSTRACT

Mushrooms represent a tremendous source of biologically useful and pharmacologically active molecules. Recent breakthroughs in cancer genetics, genomics, proteomics and translational research have helped us to develop a better understanding of the underlying mechanisms which are contributory in cancer development and progression. Different signaling pathways particularly, Wnt, SHH, TGF/SMAD and JAK/STAT have been shown to modulate cancer progression and development. Increasingly it is being realized that genetic/epigenetic mutations and loss of apoptosis also mandate a 'multi-molecular' perspective for the development of therapies to treat cancer. In this review we attempted to provide an overview of the regulation of different signaling pathways by mushrooms and their bioactive compounds. Regulation of Wnt and JAK-STAT pathways by mushrooms is deeply studied but we do not have comprehensive information about regulation of TGF/SMAD, Notch and TRAIL induced signaling pathways because of superficially available data. There are outstanding questions related to modulation of oncogenic and tumor suppressor microRNAs by mushrooms in different cancers. Therefore, detailed mechanistic insights related to targeting of multiple pathways by extracts or bioactive compounds from mushrooms will be helpful in bridging our current knowledge gaps and translation of medicinally precious bioactive molecules to clinically effective therapeutics.

## 1. Introduction

Traditional knowledge is gradually being implemented in medicine and we have witnessed landmark findings which have revolutionized treatment strategies for cancer. Substantial fraction of information has been added into the existing pool of knowledge related to dysregulated intracellular signaling cascades in different cancers. Technological advancements such as techniques for production of secondary plant metabolites, high-throughput screening and combinatorial synthesis have helped us to explore true potential of natural products in regulation of signaling pathways in different cancers (Pistollato et al., 2015; Vanden Berghe, 2012).

Mushrooms can be defined as macro-fungi having unique fruiting bodies which are either hypogeous (underground fruiting bodies) or epigeous (fruiting bodies above the ground). It is becoming progressively more understandable that extracts and bioactive components isolated from different mushrooms have strong anti-cancer activities (Jiang et al., 2014; Money, 2016). Journey started in 1970s, when different research-groups worked on pharmacological activities of mushrooms and opened new horizons for the discovery of therapeutically efficient molecules to treat cancer. *Lentinus edodes* (Berk.) (Chihara et al., 1970) and spores of *Agaricus bisporus* (Vogel et al., 1975) were noted to be effective against cancer cells. Medicinal importance of mushrooms can be well judged by entry of different mushrooms extracts and bioactive molecules in various phases of clinical trials. Moreover, cellular studies have also unveiled regulation of signaling pathways by different extracts and bioactive chemicals from mushrooms (**shown in** Table 1). In this review we set spotlight on (1) regulation of Wnt signaling, (2) targeting of JAK-STAT pathway, (3) targeting of EGFR and VEGFR Pathways, (4) regulation of mTOR-driven pathway, (5) regulation of FOXO3 and its target proteins, (6) regulation of TRAIL-induced apoptosis, (7) regulation of different proteins by bioactive molecules isolated from mushrooms, and lastly we discuss the findings related to clinical trials. Following section deals with

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**Table 1**
Cellular studies on anti-cancer effects of different extracts and bioactive chemicals from mushrooms.

| Mushroom/Bioactive Constituents | Cancer Cells | Target Proteins Activated/Increased | Target Proteins Inactivated/ Reduced | Reference |
|---|---|---|---|---|
| *Antrodia cinnamomea* | 786–0 (Renal Cell Carcinoma) | Tissue inhibitors of metalloproteinases (TIMPs) | FAK, Src kinase, Paxillin, MMP-7, ERK1/2 | Chiu et al., 2017 |
| *Antrodia camphorata* | SW480(claudin-1$^-$), SW620(claudin-1$^+$) Colon Cancer | | Claudin-1, MMP-2, MMP-9, PI3K/AKT/NFκB | Hseu et al., 2017 |
| *Ganoderma applanatum* | Caco-2 (Colon Cancer) | Bax/Bcl-2 Ratio, Caspase-3 | | Elkhateeb et al., 2018 |
| *Ganoderma lucidum* | HCT116 (Colon Cancer) | E-cadherin | MMP-1, MMP-2 | Li et al., 2017 |
| *Ganoderma resinaceum*(α-spinasterol) | MCF-7 (Breast Cancer) | p53 and Bax | Cyclin-dependent Kinase (CDK4/6) | Sedky et al., 2018 |
| *Polysaccharide from Hericium erinaceus* | GES-1 (Gastric Cancer) | Bax, Caspase-3 | Bcl-2 | Wang et al., 2017 |
| *Inonotus taiwanensis* (Polysaccharide extract) | THP-1 and U937 (Acute Monocytic Leukemia) | Caspase-9, caspase-3, PARP (poly(ADP-ribose) polymerase) cleavage γ-H2AX | LC3-II | Chao et al., 2018 |
| *Lentinus edodes* (Agaricomycetes) | HEp-2 (Larynx Cancer) | ROS production, Mitochondrial depolarization | | Finimundy et al., 2018a, b |
| *Phellinus igniarius* (Ethanolic extract) | SGC-7901 (Gastric Cancer) | Caspase-9, -3 Bax/Bcl-2 Ratio | Cyclin D1 | Wang et al., 2018a, b |
| *Pleurotus eryngii* | HCT116 (Colon Cancer) | p21, p53, caspase-3 | Cyclin B, Cyclin E and CDC-2 | Yuan et al., 2017 |
| *Pleurotus eryngii (Polyphenols-rich extract)* | HCT116 (Colon Cancer) | Caspase-3, Cleaved caspase-3. | Cyclin B, Cyclin E | Hu et al., 2018 |
| *Pleurotus sajor*-caju | HCT116 (Colon Cancer) | p53, Bax, Caspase-3 | Bcl-2 | Finimundy et al., 2018a, b |
| *Sarcodon aspratus* (Polysaccharide) | HeLa Cells | Cytochrome c, Bax, Caspase-3 | Bcl-2 | Wang et al., 2018a, b |

regulation of Wnt signaling by different mushrooms in various cancers.

## 2. Regulation of Wnt signaling

In absence of signaling, Axin and adenomatous polyposis (APC) brought *β*-catenin to glycogen synthase kinase-3β (GSK-3β) and casein kinase-1. *β*-Catenin was consequently phosphorylated by these kinases at specifically located serine/threonine residues. Phosphorylated-*β*-catenin was polyubiquitinated by B-TrCP and proteasomally degraded. Upon activation by Wnt, Frizzled receptors recruited cytosolically located Dishevelled (DVL) proteins to cell membrane. Dishevelled recruited the Axin–GSK3β complex, which phosphorylated Pro-Pro-Ser-Pro motif/s in cytosolically located segment of LRP5 (low-density lipoprotein receptor-related protein) or LRP6 (shown in Fig. 1).

WNT3a dose-dependently increased the phosphorylated levels of LRP6 and DVL2 (Zhang, 2017). Mobility of phosphorylated-DVL2 was slower on SDS-PAGE gel as detected by DVL2 antibody. *Ganoderma lucidum* treatment significantly decreased the phosphorylated-LRP6 and DVL2 in treated cells (shown in Fig. 1). Activation of Wnt pathway induced transcriptional upregulation of Axin2, c-Myc and cyclinD1 in MDA-MB-231 and 4T1 cells. *G. lucidum* (Reishi) markedly down-regulated WNT3a-activated Axin2 expression (Zhang, 2017).

4-Acetyl-antroquinonol B (4-AAQB), a biologically active chemical isolated from mycelia of *Antrodia camphorata* was noted to be effective against colorectal cancer. 4-AAQB potentially disrupted LGR5–Wnt/β-catenin pathway and dose-dependently down-regulated mRNA and protein levels of both LGR5 (leucine-rich repeat-containing G-Protein-coupled receptor 5) and *β*-catenin signaling pathway (Chang et al.,

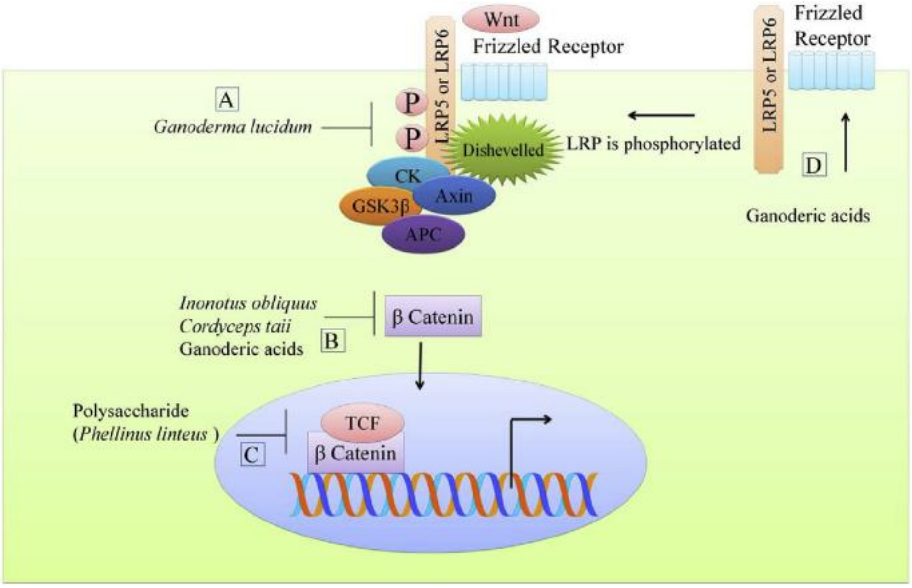


**Fig. 1.** Shows regulation of Wnt signaling by different mushrooms and their bioactive chemicals. (A) *G. lucidum* inhibited phosphorylation of LRP6. (B) *Inonotus obliquus, Cordycep staii* and ganoderic acids effectively inhibited *β*-catenin. (C) Polysaccharide isolated from *Phellinus linteus* markedly impaired association of *β*-catenin with transcriptional machinery to repress expression of target genes. (D) Ganoderic acids have been noted to increase the levels of LRP and Frizzled receptors.

2015).

*Cordycep staii* is an edible mushroom of medicinal importance and highly appreciated as an unparalleled source of biologically active chemicals (Xiao et al., 2017). Helvolic acid isolated from *C. staii* was noted to effectively inhibit Wnt signaling pathway. There was a significant reduction in the levels of B-catenin, PCNA and cyclin D1 in tumor tissues of xenografted mice combinatorially treated with helvolic acid and cyclophosphamide (Xiao et al., 2017). Ganoderic acids repressed expression levels of WNT5α/β and β-catenin and enhanced phosphorylated-GSK3β at 9th serine residue in CH157MN and IOMM-Lee cells (Das et al., 2015). Ganoderic acids down-regulated TSPAN12 (Tetraspanin) expression and simultaneously up-regulated FZD4 and its co-receptor LRP5 in CH157MN and IOMM-Lee cells (shown in Fig. 1) (Das et al., 2015).

Polysaccharide isolated from *P. linteus* demonstrated immunomodulatory activity. Polysaccharide dose-dependently repressed *β*-catenin and cyclin D1 levels in SW480 cells (Song et al., 2011). *β*-Catenin transcriptionally regulated target genes by formation of a multi-protein complex with TCF/LEF transcription factor family and binding to the promoter regions of different target genes. Luciferase reporter construct controlled by TCF/LEF response element was transiently transfected into the cells to note the effects of polysaccharides. Polysaccharide dose-dependently inhibited TCF/LEF reporter activity in SW480 cells (shown in Fig. 1) (Song et al., 2011).

Ergosterol peroxide from *Inonotus obliquus* (chaga mushroom) completely suppressed the levels of *β*-catenin in nuclear lysates of HT-29 and HCT116 cells at 5 μg/mL and 20 μg/mL, respectively. Ergosterol peroxide dose-dependently inhibited the levels of c-Myc, cyclin-dependent protein kinase 8 (CDK8) and cyclin D1 (*β*-catenin target genes) in colorectal cancer cells (Kang et al., 2015).

## 3. JAK/STAT pathway

Cytokine-induced Janus kinase (JAK)–signal transducer and activator of transcription (STAT) signaling cascade played decisive role in cancer development and progression (Shuai and Liu, 2003). Signaling strengths, specificity and kinetics of the JAK–STAT signaling cascade were regulated at multiple levels by discrete regulatory proteins (Shuai and Liu, 2003).

4-Acetyl-antroquinonol B (4-AAQB), a biologically active chemical isolated from mycelia of *Antrodia camphorata* was noted to be effective against colorectal cancer. 4-AAQB decreased phosphorylated-JAK2 and phosphorylated-STAT3 in the colonospheres and subsequently suppressed expression of target genes of JAK–STAT pathway, including, p-Src, LGR5, TYK2 and Nanog (Chang et al., 2015).

Antrocin, a sesquiterpene lactone isolated from *A. camphorate*, effectively and dose-dependently inhibited constitutively phosphorylated STAT3 at Tyr705 site (Yeh et al., 2013). p-TYK, p-Akt (Ser473), p-JAK1 (Try1022) and p-JAK2 (Try1007) were also significantly inhibited after treatment with 5 μM and 10 μM antrocin. Antrocin also impressively inhibited IL-6 mediated nuclear accumulation of STAT3 and STAT3-DNA-binding activity. Significant decrease in the activity of STAT3 was noted in let-7c overexpressing H441 and IL-6-induced PC9 cells (Yeh et al., 2013). Let-7c binding site was identified in 3′UTR of IL-6R mRNA which verified IL-6R as a target of let-7c. Markedly reduced levels of p-ERK were noted in STAT3 depleted cells. Furthermore, Mcl-1 was down-regulated after treatment with antrocin in STAT3 depleted cells (Yeh et al., 2013).

Active hexose correlated compound (AHCC) obtained from basidiomycete mushroom cultures was found to be effective against ovarian cancer cells (Choi et al., 2018). AHCC time-dependently reduced phosphorylated-STAT3 levels in CAOV3 and OV90 cells. AHCC exerted repressive effects on STAT3 activity by inducing the expression of SHP-1 in ovarian cancer cells (Choi et al., 2018). In the upcoming section, we will discuss how EGF/EGFR and VEGF/VEGFR signaling pathways can be therapeutically exploited.

## 4. Targeting of EGFR and VEGFR pathways

Growth factors transduced the signals intracellularly to regulate different developmental and cellular mechanisms. However, deregulation of these pathways played viral role in carcinogenesis. Epidermal growth factor (EGF/EGFR) and vascular endothelial growth factor (VEGF/VEGFR) signaling pathways have gained considerable attention and different chemical inhibitors designed against these pathways have entered into various phases of clinical trials.

Ling Zhi-8 (LZ-8), obtained from *Ganoderma lucidum* has recently been reported to effectively inhibit epidermal growth factor receptor (EGFR)-mediated intracellular signaling in non-small-cell lung cancer cells (NSCLC) (Lin et al., 2017). LZ-8 abolished EGFR autophosphorylation at the 1068th and 1173rd tyrosine residues after 3 h and 24 h. Moreover, EGFR dimerization assay indicated that LZ-8 inhibited EGFR dimerization. LZ-8 and EGF induced scattered distribution of EGFR and consequently appeared as punctated intracellular clusters in the cytoplasm in H226 and A549 cells. LZ-8-mediated degradation of EGFR was noted to be dependent on clathrin-mediated endocytosis in NSCLC cells (Lin et al., 2017). Significantly reduced turnover of LZ-8-induced EGFR degradation was noted in clathrin silenced NSCLC cells. LZ-8 promoted EGFR ubiquitination and degradation by activation of EGFR degradation pathway. LZ-8 induced significantly higher binding of c-Cbl to EGFR in A549 and H226 cells. EGFR levels were considerably enhanced in c-Cbl silenced A549 and H226 cells (Lin et al., 2017).

*Antrodia cinnamomea* ethanolic extract (ACEE) dose dependently reduced VEGFR2 in lung cancer LLC cells (Huang et al., 2018). ACEE produced anti-angiogenic effects by inhibition of expression of VEGFR2 in LLC cells. Findings obtained from immunohistochemistry assay clearly indicated that VEGFR2 was reduced in allograft lung metastases produced by injection of LLC cells into the right hind paw of mice. CD31, an indicator of micro-vessel density was down-regulated in mice treated with ACEE (Huang et al., 2018). Based on these findings, it can be concluded that mushrooms offer a good reservoir of biologically active molecules to effectively target EGFR and VEGFR triggered pathways. In the next section, we will focus on targeting of mTOR-driven pathway.

## 5. Regulation of mTOR-driven pathway

Mechanistic target of rapamycin (mTOR) has attracted substantial attention because of its hallmark feature of regulation of mechanism of protein synthesis. mTOR worked synchronously with different proteins and modulated various downstream target proteins. Eukaryotic initiation factor cap-initiation complex (eIF4G, eIF4E, eIF4A) interacted with poly-adenylation tail binding protein (PABP) at the 3′ poly-adenylated end of mRNAs. Functionally active mTORC1 maintained cap-dependent protein synthesis by phosphorylation and inactivation of 4E-BPs and sequestered it away from eIF4E. Active mTORC1 phosphorylated and activated ribosomal S6 kinases (S6Ks), which phosphorylated eIF4B and increased its binding to eIF4A.

Severe combined immunodeficient mice (SCID) injected with inflammatory breast cancer cells treated with Reishi showed reduced tumor growth reduced (Suarez-Arroyo et al., 2013). mTOR, p70S6K and eIF4G were found to be significantly reduced in xenografted mice treated with Reishi (Suarez-Arroyo et al., 2013). Grifolin, a secondary metabolite obtained from *Albatrellus confluens* markedly reduced phosphorylated levels of Akt/PKB, mTOR, p70S6K, S6 and 4E-BP1 in ovarian cancer A2780 and SKOV3 cells (Che et al., 2016). Shown in Fig. 2.

*Pleurotus nebrodensis* polysaccharide (PN50G) markedly inhibited phosphorylated levels of Akt, mTOR and p70S6K in lung cancer A549 cells (Fig. 2) (Cui et al., 2016).

Antroquinonol isolated from *Antrodia camphorata* significantly inhibited phosphorylation of p70S6K at 389$^{th}$, 421$^{st}$ threonine residues

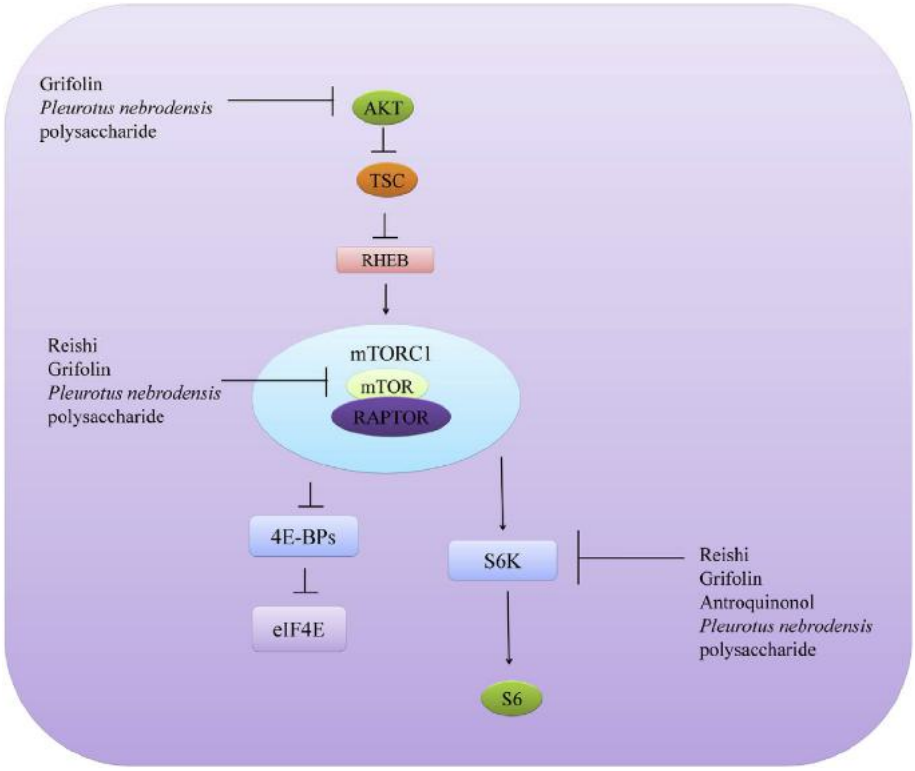


**Fig. 2.** Schematically represents targeting of upstream activators and downstream effectors of mTOR. AKT, mTOR and S6K were inhibited by Grifolin and *Pleurotus nebrodensis* polysaccharide. Protein kinase B (PKB/AKT), S-6 kinase (S6K), mTOR Complex 1 (mTORC1), eIF4E-binding proteins (4E-BPs).

and serine 424, eIF4E at 209$^{th}$ serine residue and 4E-BP1 at 37th, 46th and 70$^{th}$ threonine residues in pancreatic cancer AsPC-1 cells (Yu et al., 2012). It seems exciting to note that bioactive molecules isolated from different mushrooms have shown remarkable efficacy against mTOR-driven pathway. However, we still have to identify the biologically active molecules which can target mTORC1 and mTORC2 in cancer cells.

## 6. Regulation of FOXO3 and its target proteins

Forkhead box O (FOXO) transcriptional factors are essentially involved in the regulation of apoptosis and cell cycle (Eijkelenboom and Burgering, 2013). Multiple upstream pathways regulated activity of FOXO proteins mainly through post-translational modifications (phosphorylation, ubiquitylation and acetylation) and nuclear–cytoplasmic transportation of FOXO proteins. These proteins moved into the nucleus and transcriptionally up-regulated pro-apoptotic genes (Eijkelenboom and Burgering, 2013).

Ergosterol isolated from *Amauroderma rude* induced significantly longer survival time in tumor bearing mice. Mean survival duration of ergosterol-treated mice was 10.9 weeks (ranged 3–15 weeks) (Li et al., 2015a; b). After 15 weeks, 5 mice from the ergosterol-treated group survived and had better health conditions. Ergosterol increased mRNA and protein levels of FOXO3 in MDA-MB-231 BCa cells. FOXO3 was noted to stimulate the expression of Fas and FasL in treated breast cancer cells. Ergosterol significantly enhanced Fas and FasL (Fas ligand) in treated breast cancer cells (Li et al., 2015a; b). Circumstantial evidence also verified anticancer activity of ergosterol peroxide against HepG2 cells (Li et al., 2016). BAX and PUMA are pro-apoptotic proteins and noted to be controlled by FOXO3. BAX and PUMA were notably enhanced in ergosterol peroxide treated HepG2 cells (Li et al., 2016).

## 7. SHH pathway

Certain hints have emerged which suggested that 4-AAQB dose-dependently down-regulated smoothened (SMO) mRNA expression and consequently disrupted SMO-dependent SHH pathway (Chang et al., 2015).

## 8. TGF/SMAD signaling

Transforming growth factor (TGF) transduced the signals intracellularly through specific receptors (TGFR). Phosphorylation of activation of Mothers against Decapentaplegic (SMAD) promoted its entry into the nucleus to transcriptionally regulate expression of target genes.

Sulfated polysaccharide (SPS) obtained from *Antrodia cinnamomea* dose-dependently reduced phosphorylated-SMAD2/3 in LLC1 and A549 cells (Lu et al., 2017). SPS decreased TGF1-induced phosphorylation of SMAD2/3, PKB/AKT and FAK. SPS dose-dependently down-regulated TGFRII and TGFRI protein levels. SPS induced degradation of TGFR relied on ubiquitin (Ub)–proteasome pathway (UPP)-mediated degradation. Expectedly, SPS (200 μg/mL) treatment resulted in 50% and 40% reduction in the TGFRI proteins in LLC1 and A549 cells, respectively. Moreover, use of MG-132 (proteasome inhibitor) chaperoned TGFRs from SPS-induced protein degradation (Lu et al., 2017).

It is evident that mushroom derived molecules have the ability to regulate TGF/SMAD pathway. However, we still have incomplete information about the exact mechanism. It needs to be tested in different cancers and more importantly the mechanism used by these chemicals to inhibit TGF/SMAD pathway. Seemingly, these chemicals inhibited phosphorylation of transducer proteins. But it needs to be investigated it mushroom-derived products also induced degradation of these proteins or activated negative regulators of TGF signaling.

## 9. microRNA regulation

Discovery of microRNAs has paradigmatically shifted our previous concepts related to "central dogma of molecular biology". We now know that miRNAs are non-coding RNAs having the ability to target mRNAs of myriad of genes. Excitingly, mushrooms and their bioactive molecules have been shown to regulate the expression of different miRNAs in cancers.

Infiltration of Regulatory-T cells ($T_{regs}$) played an important role in tumor immunoevasion and blocked effective HCC immunotherapy (Li et al., 2015a; b). *Ganoderma lucidum* polysaccharides (GLPS) effectively reduced intratumoral accumulation of $T_{regs}$. However, ratio of effector T-cells was dramatically enhanced. GLPS exerted inhibitory effects on $CD4+CD25^{+}$ Treg cells. Forkhead box P3 (FOXP3) was found to be a master transcriptional factor that regulated development and function of T-cells. NOTCH signaling controlled FOXP3 through Hes1 and RBP-J-dependent mechanisms. Structural studies have verified presence of RBP-J-binding site and N-box sites within promoter region of FOXP3 (Li et al., 2015a; b). GLPS triggered miRNA-125b directly targeted FOXP3 and NOTCH1 in T cells (Li et al., 2015a; b).

*Ganoderma lucidum* extracts significantly down-regulated hsa-miR-27a* (4.46-fold) and up-regulated hsa-miR-1285 (10.46-fold) in breast cancer cells (Gonul et al., 2015). Telomerase catalytic subunit (TERT) in association with the telomerase RNA component (TERC) formed functionally active enzyme. Many microRNAs have been shown to quantitatively control catalytic subunit of the hTERT. miR-1207–5p (up-regulated 2.895-fold) and miR-3687 (up-regulated 2.15-fold) were also noted to be changed (Gonul et al., 2015).

Polysaccharides from *G. lucidum* up-regulated miR-3131 in HepG2 cells (Shen et al., 2014). However, the team did not provide any convincing evidence related to oncogenic or tumor suppressive role of miR-3131 in HepG2 cells. Although there is a recent report which provided evidence that over-expression of miR-3131 inhibited apoptosis in SK-Hep-1 cells (Wang et al., 2017). Although existing literature is not substantial; however, future studies must unlock full potential of mushroom-derived molecules as versatile regulators of different miRNAs in various cancers.

## 10. TRAIL induced apoptosis

TRAIL had emerged as a premium anticancer agent because of its selective targeting ability of cancer cells. However, contemporary studies also identified development of resistance against TRAIL in different cancers. Proteomic studies suggested that imbalance of pro- and anti-apoptotic proteins and down-regulation of death receptors severely impaired TRAIL-driven apoptosis. cIAP-1, cIAP-2, Mcl-1 and XIAP are frequently over-expressed anti-apoptotic proteins in cancers. Whereas, Bax and Bak are pro-apoptotic proteins which promote TRAIL induced apoptosis. TRAIL transduced the signals intracellularly through death receptors (DR4 and DR5).

Clitocine, a naturally occurring nucleoside isolated from wild mushroom has been shown to sensitize colon cancer cells to TRAIL induced apoptosis (Sun et al., 2016). Clitocine interfered with the binding between Bak and Mcl-1 and promoted mitochondrial translocation of Bax which resulted in mitochondrial membrane permeabilization. Mcl-1 over-expression efficiently attenuated clitocine and TRAIL mediated apoptotic cell death by suppression of intrinsic pathway (Sun et al., 2016).

Pachymic acid, a lanostane type triterpenoid obtained from *Poria cocos* dose-dependently up-regulated DR5 and TRAIL; however, the levels of DR4, Fas and FasL remained unchanged in bladder cancer cells (Jeong et al., 2015). Pachymic acid up-regulated proapoptotic proteins (Bad and Bax) and down-regulated anti-apoptotic proteins (Bcl-xL and Bcl-2). Pachymic acid also dose-dependently suppressed cIAP-1, cIAP-2 and XIAP in bladder cancer cells (Jeong et al., 2015).

Peptides obtained from *Lentinus squarrosulus* reduced viability of lung cancer H460 cells up to 40% when used at a concentration of 20 μg/mL (Prateep et al., 2017). 85% reduction in cellular viability was noted at a concentration of 40 μg/mL. Peptides efficiently down-regulated Bcl-2 and FLICE inhibitory protein (c-FLIP) in lung cancer cells. Level of functionally active caspase-8 was noted to be enhanced in peptides treated cancer cells (Prateep et al., 2017). Fermented culture broth of *Antrodia salmonea* dose-dependently cleaved procaspase-8 into its active form (Chang et al., 2017). *A. salmonea* significantly upregulated FasL and Fas receptor in MDA-MB-231 BCa cells. *A. salmonea* strongly inhibited tumor growth in nude mice intravenously injected with MDA-MB-231 BCa cells (Chang et al., 2017).

## 11. Regulation of different proteins by bioactive molecules isolated from mushrooms

It seems exciting to note that mushrooms offer a huge resource of biologically and pharmacologically active chemicals. Because of rapidly developing resistance against various drugs, it is imperative to identify natural products having significant activity and minimum off-target effects.

### 11.1. Grifolin

Grifolin, a secondary metabolite isolated from mushroom *Albatrellus confluens* was noted to be pharmacologically active against different cancer cell lines (Luo et al., 2016). Grifolin markedly reduced migratory capacity of metastatically competent cancer 5–8F and MGC803 cells. Grifolin significantly repressed expression and activity of CD44 and MMP2 (matrix metalloproteinase) in 5–8F and MGC803 cells. Secreted MMP2 significantly decreased with an inhibition rate of 77.2% and 41.8% in MGC803 and 5–8F cells after treatment with grifolin. Grifolin considerably down-regulated PGC1α (PPAR-Gamma co-activator) in MGC803 and 5–8F cells. There was a significant reduction in MMP2 and CD44 levels in PGC1α depleted cells, whereas, levels of MMP2 and CD44 increased substantially in PGC1α over-expressing non-metastatic nasopharyngeal carcinoma 6–10B cells (Luo et al., 2016). Co-immunoprecipitation provided evidence of existence of a protein-protein interaction between PGC1α and transcriptional factors, activator protein (AP-1) (Fra-1/c-Jun) and LSF. Endogenously occurring PGC1α co-immunoprecipitated with LSF or Fra-1 in MGC803 and 5–8F cells, respectively. There were significantly impaired structural associations of PGC1α/Fra-1 and PGC1α/LSF in grifolin treated cancer cells. Immunofluorescence staining revealed that grifolin exerted repressive effects on nuclear colocalization of PGC1α with LSF or Fra-1 in MGC803 and 5–8F cells, respectively. It has recently been convincingly revealed that grifolin interfered with protein-protein interactions of PGC1α/Fra-1 and PGC1α/LSF to down-regulate CD44 and MMP2 (Luo et al., 2016). Grifolin structurally interacted with ERK1/2 and exerted inhibitory effects on the activities of ERK1/2 in CNE1 nasopharyngeal carcinoma cells, HeLa cervical cancer cells and MCF7 breast cancer cells (Luo et al., 2015). Grifolin physically interacted with the ATP pocket of the ERK2 (extracellular signal-regulated kinase). Grifolin (32 mg/kg/day) for 25 days significantly decreased metastasis up to 18.2% in BABL/c nude mice injected with highly metastatic nasopharyngeal carcinoma 5–8F cells. Adverse-effects-associated with grifolin treatment were not observed in treated mice. Data clearly suggested that the metastasizing activity of 5–8F-Z cells to the lung was significantly reduced in grifolin treated xenografted mice. ELK1, a member of ETS oncogene family is a downstream effector of ERK1/2 signaling (Luo et al., 2015). Grifolin decreased the phosphorylation of ELK1 (S383) in 5–8F and MDA-MB-231 cells. Treatment with grifolin dose-dependently down-regulated DNMT1 in cancer cells. Several potential binding sites for ELK1 have been identified in proximal promoter region of DNMT1. Grifolin dose-dependently down-regulated mRNA/protein expression of DNMT1 (Luo et al., 2015).

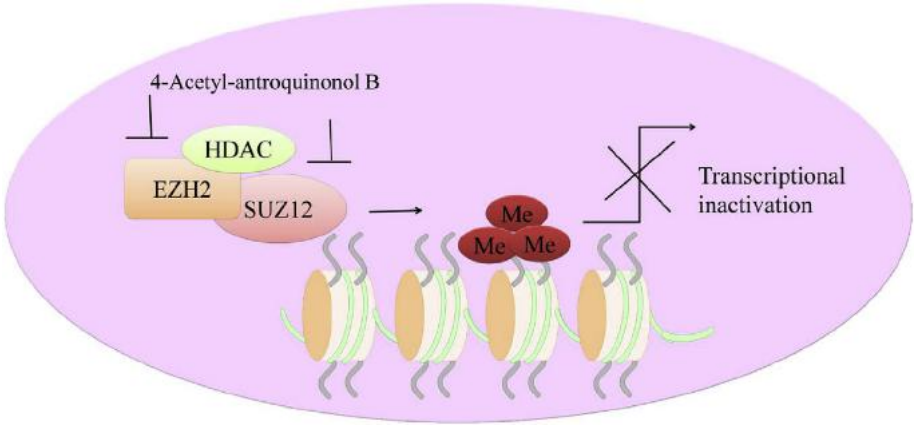


**Fig. 3.** Schematically represents epigenetic inactivation of genes by positioning of EZH2, SUZ12 and HDAC. 4-AAQB inhibited EZH2 and SUZ12.

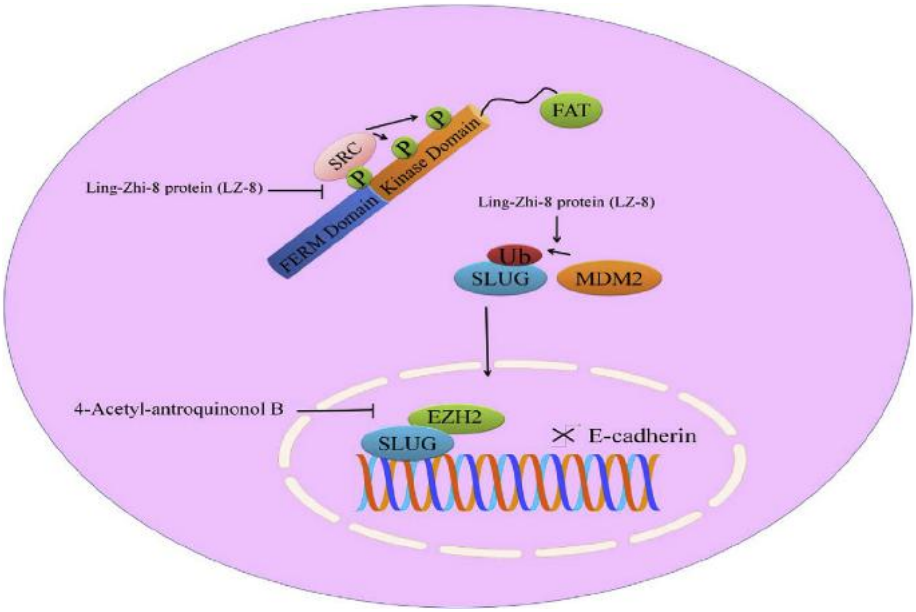


**Fig. 4.** Shows activation of FAK and association of SRC with active FAK. SLUG transcriptionally downregulated E-cadherin and stimulated metastasis associated genes. MDM2 targeted SLUG for degradation.

### *11.2. 4-Acetyl-antroquinonol B (4-AAQB)*

4-Acetyl-antroquinonol B (4-AAQB), a biologically active compound isolated from mycelia of *Antrodia camphorata* was noted to be effective against colorectal cancer. 4-AAQB considerably reduced viability of CD133+ DLD-1 cells (Chang et al., 2015). It has been noted that 5-fluorouracil and oxaliplatin increased the number of ALDH + colorectal cancer stem cells (CRC-CSCs) by 18.7% and 10.7%, respectively. Interestingly, 4-AAQB dose-dependently increased the sensitivity of 5-fluorouracil-resistant CRC-CSCs (Chang et al., 2015).

Progress over past decade had demystified two main multi-protein assemblies, Polycomb repressive complex 1 (PRC1) and PRC2 which played instrumental role in epigenetic inactivation (Simon and Kingston, 2009). PRC2 comprised different core proteins: EZH2 (enhancer of Zeste-2) and SUZ12 (suppressor of Zeste-12), along with its accessory proteins (shown in Fig. 3). This multi-protein complex catalyzed methylation of K27 in the histone H3 tail. 4-AAQB exerted inhibitory effects on tumor-promoting epigenetic factors EZH2 and SUZ12 (Fig. 3) (Chang et al., 2015).

Together with the rapidly growing understanding about intricate interplay between PRC1 and PRC2 than what was previously surmised, it is essential to therapeutically target proteins of these complexes more effectively.

### *11.3. Ling-Zhi-8 protein (LZ-8)*

SNAIL/SLUG are multifaceted transcriptional factors which play contributory role in epithelial–mesenchymal transition (EMT) (Lamouille et al., 2014). SLUG is reportedly involved in transcriptional downregulation of epithelial marker genes and upregulation of genes correlated with mesenchymal phenotype (Lamouille et al., 2014). SLUG triggered positioning of the polycomb repressive complex (PRC) at promoter region of E-cadherin to repress it transcriptionally. Enhancer of zeste homologue 2 (EZH2) was an essential protein of PRC (Anzai et al., 2017; Xu et al., 2017).

Ling-Zhi-8 protein (LZ-8) inhibited the colonization of tumor cells at distant sites in lungs and also markedly reduced metastasizing potential of lung cancer cells (Lin and Hsu, 2016). LZ-8 remarkably repressed tumor weight and volume in C57BL/6 mice subcutaneously injected with LLC1 cells. LZ-8 induced epithelial-like morphological characteristics, particularly up-regulated γ-catenin and E-cadherin, as well as down-regulated mesenchymal-like markers vimentin and N-cadherin. LZ-8 promoted re-localization of vinculin and morphological features of CL1-5 and A549 cells changed from filopodia shaped into round-shaped

after treatment with LZ-8 (Lin and Hsu, 2016).

Phosphorylation of focal adhesion kinase (FAK) at 397$^{th}$ tyrosine residue created a motif that was used as a scaffold by different SH2-domain containing proteins. SRC induced phosphorylation of the FAK kinase domain (Y576 and Y577) (shown in Fig. 4). LZ-8 inhibited FAK autophosphorylation and simultaneously inhibited Src phosphorylation in lung cancer cells (Fig. 4). Data clearly suggested that LZ-8 bound to filopodia and interfered with organization of focal adhesion signaling complexes by inhibiting FAK-induced transduction cascade and cell mobility (Lin and Hsu, 2016). LZ-8 blocked FAK activation and inhibited FAK-mediated SLUG expression. LZ-8 time-dependently enhanced SLUG ubiquitination by promoting association between MDM2 (mouse double minute 2) and SLUG (Fig. 4). SLUG levels elevated significantly in MDM2 silenced cancer cells and interestingly, knockdown of MDM2 severely impaired LZ-8-mediated suppression of SLUG (Lin and Hsu, 2016).

## 12. Xenografted mice studies

Ostreolysin isolated from *Pleurotus ostreatus* has been shown to be effective against colon cancer cells. Recombinant ostreolysin was used and results revealed that cells lost their elongated shape and displayed higher levels of cleaved executioner caspases-3 and 7 (Nimri et al., 2017). Moreover, cytosolic accumulation of droplet-like structures of different sizes was also observed in treated cancer cells. Recombinant ostreolysin considerably reduced tumor growth in mice xenografted with colon cancer cells (Nimri et al., 2017).

Intraperitoneally injected metronomic zoledronic acid (mZOL) (0.0125 mg/kg x 8times over 4 weeks) effectively reduced size of the tumor and markedly inhibited lung and liver metastasis in xenografted mice (Ko et al., 2017). *Trametes versicolor* was combined with mZOL to unravel synergistic effects and data verified that both agents combinatorially reduced tumor-induced bone destruction in tumor-bearing tibia and considerably inhibited metastasis to both lung and liver (Ko et al., 2017).

## 13. Clinical trials

*Agaricus blazeii* Murrill (AbM), a Basidiomycetes mushroom has been tested for clinical efficacy. AndoSan contained 82% of AbM together with two other mushrooms (*Hericium erinaceus* and *Grifola frondosa*) (Tangen et al., 2015). Notable increase in interleukin-1 receptor antagonist (IL-1RA), IL-5, and IL-7 serum levels was found in the Agaricus group. Elevated levels of IL-7 found in the AndoSan group might be considered as a positive treatment effect. Plasmacytoid dendritic cells (pDC) stimulated both adaptive and innate immune system and induced tolerance. However, frequently reduced levels of pDC are noted in multiple myeloma patients (Tangen et al., 2015). However, AndoSan induced an increase in the population of pDC which was a positive treatment effect in these patients. Immunoglobulin related genes (IGKC, IGKC and IgHV4-31) and genes related to natural killer (NK) cells, killer immunoglobulin receptors (KIR2DL4 and KIR2DL3) were up-regulated in the Agaricus group. Future studies must converge on a detailed analysis of multiple myeloma patients undergoing high-dose chemotherapeutics and autologous stem cell transplantation (Tangen et al., 2015).

Plasma levels of IFN-γ and IL-12 were noted to be increased after oyster mushroom (*Pleurotus cornucopiae*) consumption (Tanaka et al., 2015). Moreover, IFN-γ, a strong stimulator of NK cells, increased significantly by oyster mushroom intake compared to the placebo at week 8. Oyster mushroom extract stimulated Th1-type cytokines release; however, Th2-type cytokines were minimally influenced. Oyster mushroom extract consumption by humans up-regulated IFN-γ and consequently boosted immune system (Tanaka et al., 2015).

In a recently reported trial, *Antrodia cinnamomea* was given in combination with chemotherapeutic drugs to cancer patients (Tsai et al., 2016). Although, a lower mortality rate and longer mean overall survival in *A. cinnamomea* treated group was observed but co-treatment with *A. cinnamomea* and chemotherapy did not improve the outcome of advanced cancer patients. Instead, combinatorial treatment lowered platelet counts (Tsai et al., 2016).

Thirty six prostate cancer patients were given tablets of *Agaricus bisporus* and dose limiting toxicities were not noticed. Overall, PSA (prostate specific antigen) response rate was 11% (Twardowski et al., 2015). Two patients who received 8 and 14 g/d completely responded to the treatment. PSA levels were lowered to almost undetectable values in patients who continued for 49 and 30 months. Two patients who received 8 and 12 g/d partially responded to the treatment. After 3 months of therapy, decline in the levels of PSA below baseline was recorded in 13 patients. Complete-responders and partial-responders presented high values of baseline IL-15 (interleukin-15) as compared to non-responders. Therapy-associated declines in MDSCs (myeloid-derived suppressor cells) were also recorded in this group (Twardowski et al., 2015).

First course of treatment consisted of only chemotherapeutic drugs and 2nd course consisted of concomitantly administered *Lentinula edodes* mycelia extract and chemotherapeutic drugs (Yamaguchi et al., 2011). In comparison to the pre-chemotherapy state, changes were not noticed in immune parameters or quality of life after 1st course of chemotherapeutic drugs. However, following second course of combinatorial treatment, improved clinical findings were noted related to quality of life, immunosuppressive acidic protein (IAP) levels and NK cell activity (Yamaguchi et al., 2011).

Active hexose correlated compound (AHCC), a culture extract of *Lentinula edodes* significantly down-regulated HHV-6 (human herpes virus) levels in saliva of cancer patients undergoing chemotherapy (Ito et al., 2014). AHCC not only improved quality of life but also chemotherapy-associated-hepatotoxicity and hematotoxicity in the cancer patients (Ito et al., 2014).

Orally administered *Trametes versicolor* preparations improved immunological responses in breast cancer patients after standard chemotherapy and radiotherapy (Torkelson et al., 2012). 9 g/day of *T. versicolor* preparation was well tolerated in breast cancer patients for 6 weeks in the post-radiation treatment settings. Immunological results pointed towards an increase in lymphocyte counts at 6 and 9 g/day and increased activity of natural killer cells at 6 g/day. Dose-related increases in CD19$^{+}$ B cells and CD8$^{+}$ T cells were noticed however, CD16+/56 + NK cells and CD4+T cells remained unchanged (Torkelson et al., 2012).

## 14. Conclusion

Overwhelmingly increasing experimentally verified data have started to shed light on regulation of multiple signaling pathways by mushrooms in different cancers. In this review we attempted to summarize most recent information about modulation of Wnt, SHH, JAK-STAT, TGF/SMAD pathways by mushrooms. On the basis of the results obtained to date and the above discussion, we propose three areas which need detailed research. (1) Since its discovery, field of microRNAs has undergone substantial broadening and we still have to witness how different oncogenic and tumor suppressor miRNAs are regulated by extracts and bioactive compounds of mushrooms. (2) TRAIL based therapies are currently being tested for efficacy in different phases of clinical trials and it will be interesting to see if combinatorial treatment consisting of TRAIL and mushrooms can efficiently inhibit cancer progression. (3) Mushroom mediated regulation of SHH and Notch signaling also needs detailed and extensive research.

## Conflicts of interest

The authors declare no conflict of interest.

## Transparency document

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